



Did the ACA Medicaid expansion save lives?☆

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ABSTRACT

We estimate the effect of the Affordable Care Act Medicaid expansion on county-level mortality in the first four years following expansion using restricted-access microdata covering all deaths in the United States. To adjust for pre-expansion differences in mortality rates between treatment and control, we use a propensity-score weighting model together with techniques from machine learning to match counties in expansion and non-expansion states. We find a reduction in all-cause mortality in ages 20 to 64 equaling 11.36 deaths per 100,000 individuals, a 3.6 percent decrease. This estimate is largely driven by reductions in mortality in counties with higher pre-expansion uninsured rates and for causes of death likely to be influenced by access to healthcare. A cost-benefit analysis shows that the improvement in welfare due to mortality responses may offset the entire net-of-transfers expenditure associated with the expansion.

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1. Introduction

Medicaid is the largest means-tested social insurance program in the U.S., providing publicly-funded health insurance for low-income families and individuals. Evidence from recent studies has shown that Medicaid reduces financial risk to beneficiaries while also increasing access to healthcare services.¹ Far less evidence links Medicaid

access to long-run improvements in health and mortality, particularly among adults.² Prior to the Affordable Care Act (ACA) expansion, estimation of the mortality effects of access to the present Medicaid program relied on a small number of state-level actions in the early to mid-2000s (Sommers et al., 2017) and suggestive but underpowered evidence from the 16-month mortality followup in the Oregon Health Insurance Experiment (OHIE) (Finkelstein et al., 2012).

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¹ There is extensive evidence pertaining to these outcomes: Currie and Gruber (1996a,b), Card and Shore-Sheppard (2004), Long et al. (2005), Finkelstein et al. (2012), DeLeire et al. (2013), Sommers et al. (2014), Taubman et al. (2014).

² The effects of the modern Medicaid program on adult health and mortality have been examined in Finkelstein et al. (2012), Baicker et al. (2013), Sommers (2017), Wherry and Miller (2019), Black et al. (2019), and Goldin et al. (2019). Goodman-Bacon (2018) shows significant mortality reductions from the initial roll out of Medicaid in 1965. We compare our estimates to the emerging literature on the effects of the ACA expansion in Section 4. The lack of health insurance is associated with worse health and higher mortality (Wilper et al., 2009), but causality is not clear (Kronick, 2009; Black et al., 2017).

In this paper, we estimate the effects of the ACA Medicaid expansion on adult mortality in the first four years following the 2014 Medicaid expansion. A key part of the ACA, the Medicaid expansion removes categorical exclusions and bases eligibility solely on income at or below 138 percent of the federal poverty level. The ACA Medicaid reform was originally formulated to occur nationwide but was effectively made a state option by the 2012 Supreme Court ruling *National Federation of Independent Business v. Sebelius*. As a result, just over half of the United States chose to adopt the initial expansion ([Center for Medicare and Medicaid Services, 2017](#)). Coverage became effective on January 1, 2014, and over 9 million new individuals between the ages of 19 and 64 enrolled by the end of 2015, accounting for 60 percent of the immediate increase in coverage that resulted from the ACA ([Frean et al., 2017](#)).

Our primary analysis compares the post-expansion changes in mortality between counties in states that expanded Medicaid in 2014 with changes in those that did not adopt the expansion. To account for pre-existing differences between these counties and improve the efficiency of our estimation, we use propensity-score reweighting based on economic, demographic, and political characteristics. Two key features of the implementation of our research design serve to discipline the analysis and add credibility to the findings. First, we use the double-lasso method to specify the propensity score model ([Belloni et al., 2014a; Urminsky et al., 2016](#)). This data-driven model-selection procedure prevents overfitting, thereby increasing power relative to research designs assessed in [Black et al. \(2019\)](#) (hereafter BHNS). Second, we perform a cross-validation exercise in which we hold out mortality outcomes in the four years before the reform from the propensity score model and check that the reweighted data displays flat pre-trends over this time period.³ Mortality responses, including age and cause of death, are observed in restricted-access microdata for all deaths in the U.S. from 1990 to 2017. Together, this allows us to estimate the marginal impact of public health insurance coverage for areas that adopted the Medicaid expansion and, further, to examine whether changes occur where Medicaid is likely to have the largest effects.

We find evidence of a reduction in mortality following the ACA Medicaid expansion, with the strongest results for amenable causes of death, those most likely to have been averted through optimal quality healthcare. Point estimates on aggregate 4-year mortality indicate an improvement in all-cause mortality among 20-to-64-year-old adults of 11.36 fewer deaths per 100,000 people, a 3.60 percent decrease in mortality. The event study shows flat pre-reform trends and an immediate drop in mortality in the first year following the reform, with evidence of growing effects after that. Changes in aggregate mortality rates are largely explained by a reduction equaling 6.64 fewer deaths per 100,000 people due to amenable causes. We also find a larger reduction in mortality in areas with low rates of health insurance coverage before the expansion.

Applying estimates of newly covered individuals in states that expanded Medicaid relative to those that did not, the aggregate effects equate to one additional death averted for each 310 new Medicaid recipients. When interpreting these magnitudes, it is important to note that access to healthcare may have larger effects over longer time horizons than available in our data, and we find growing effects over time.

An essential aspect of our study is that we examine the effect of the program on aggregate mortality, thereby capturing mortality responses for the entire population along all margins. These aggregate mortality results complement two recent papers, [Miller et al. \(2019\)](#) and [Goldin et al. \(2019\)](#), which provide evidence on the effects of the ACA for the Medicaid eligible and uninsured population. Our individual-level treatment-on-the-treated (ITOT) estimates fall within the range of estimates from this emerging literature, although they are subject to some uncertainty regarding the size of the affected population. Using our first-stage estimate of a 4.15 percent decrease in uninsurance in our sample and adjusting for the higher mortality of the Medicaid population implies that the expansion prevented 30 percent of deaths in this population, a large reduction in mortality.⁴ This ITOT extrapolation relies, however, on several strong assumptions, especially, that the beneficial effects of the expansion were confined to those measured as switching from uninsurance to Medicaid. This assumption would be violated, for example, if there were responses among those who switch onto Medicaid from other sources of insurance, Medicaid affects prospective recipients through presumptive eligibility and retroactive coverage, or if the benefits of health insurance affect households or communities. A number of program features are consistent with the first two violations, while a growing literature documents responses consistent with diffuse benefits of Medicaid on, for example, financial outcomes ([Hu et al., 2016; Miller et al., 2018](#)), crime ([Vogler, 2017; He, 2018](#)), and hospital closures ([Lindrooth et al., 2018](#)). There is also little evidence of negative spillovers through congestion effects ([Carey et al., 2018](#)). If instead we scale our estimates to the increase in Medicaid coverage regardless of previous insurance, our estimates imply a 17 percent reduction in mortality. The uncertainty in these calculations leads us to conclude that the evidence is most informative about the reduced-form, aggregate mortality effects of the expansion. We discuss these issues further in Section 4.

At the end of the paper, we examine the relationship between the estimated mortality effects and the cost-effectiveness of the ACA Medicaid expansion. Specifically, we compare program costs to the value of mortality gains using age-specific estimates of the Value of a Statistical Life (VSL) in [Aldy and Viscusi \(2008\)](#). We calculate the benefits of expansion under two sets of assumptions: first, by

³ We discuss the merits of our strategy relative to the difference-in-differences and triple difference designs assessed in BHNS in Section 3.

⁴ Large TOT estimates for the effects of public insurance coverage on mortality appear across several research designs, such as the age 64 vs. age 65 contrast in [Huh and Reif \(2017\)](#); also see [Sommers et al. \(2017\)](#) and [Dunn and Shapiro \(2019\)](#). Randomized trials for the most effective drugs, such as statins, show short-term mortality reductions around half as large ([Baigent et al., 2005; Chou et al., 2016](#)).

applying our point estimate for all-cause mortality to the ages 20 to 64 (taking our results at face value); and second, under a conservative scenario in which our results apply only to amenable cause mortality for the 55-to-64 age group (where previous researchers suggest results are most likely to be found, and our results are most robust). These results imply that the mortality benefits offset one-third of the expenditures associated with Medicaid expansion. An economic cost-benefit analysis should weigh the mortality benefits against the efficiency costs of the program, i.e. deadweight loss associated with the crowd out from private insurance and raising tax revenue. Evidence from the OHIE suggests that approximately 60 percent of Medicaid expenditures are transfers, for example, to hospitals through a reduction in unpaid bills (Finkelstein et al., 2016).⁵ Applying this figure to the ACA Medicaid expansion would imply that the mortality-related savings may cover the entire net cost of the ACA Medicaid expansion. A full accounting of the social welfare effects of the Medicaid expansion should also include benefits to other outcomes, such as quality of life improvements from changes in morbidity and reductions in financial distress among beneficiaries.

2. Data

We use information from the Kaiser Family Foundation (2018) to identify states that expanded Medicaid in the first half of 2014. We include Wisconsin as treated although their expansion was limited to adults within 100 percent of the federal poverty level. We exclude counties in six states (PA, IN, NH, AK, MT, LA) that implemented the Medicaid expansion after the first half of 2014. Results that include these states are presented as robustness checks.

Mortality data come from the National Vital Statistics System (2018) through the Centers for Disease Control and Prevention (CDC). These restricted access mortality data contain detailed information on age, sex, race, cause of death, and county of residence for every death in the U.S. The CDC assigns the cause of death by examining death certificates collected from around the country. We follow Sommers (2017) to classify amenable conditions using the ICD-10 codes included in the mortality data. Amenable conditions are presented in Appendix Table A.1. Despite the care taken by the CDC in constructing the data, it is important to note that the classification of cause of death into amenable and non-amenable may be imperfect due to difficulty in ascertaining cause of death, differences in how death certificates are filled out by coroners, and the inherent challenges in determining which causes would have been prevented by access to optimal-quality healthcare. Therefore, we use all-cause mortality as the focal outcome in our model selection procedure and report amenable cause mortality as a secondary outcome.

The mortality data are paired with population denominators by county, age group, sex, and race from the U.S. Census and then merged with county-specific economic and demographic variables, such as the unemployment rate, the poverty rate, and real median income. These data were obtained from the Small Area Unemployment Statistics (SAUS) program through the Bureau Of Labor Statistics (2018) and the Small Area Income and Poverty Estimates Program (SAIPE) through the U.S. Census (SAIPE, 2018). We also gathered county-level uninsured data from the U.S. Census Small Area Health Insurance (SAHIE) database (SAHIE, 2018).

3. Conceptual framework and empirical strategy

Theoretically, the ACA Medicaid expansion could impact mortality in either direction. On the one hand, reductions in mortality are most likely to occur through increases in healthcare use. Research has shown the Medicaid leads to increases in visits to physicians, overnight hospital stays, emergency department visits, and rates of diagnosis for medical conditions (Finkelstein et al., 2012; Taubman et al., 2014; Wherry and Miller, 2016; Duggan et al., 2019). On the other hand, Medicaid could increase mortality through increases in *ex ante* moral hazard; although, Simon et al. (2017) find no evidence that the ACA Medicaid expansions have increased moral hazard behavior. A third channel through which Medicaid could impact mortality in either direction is an income effect. Indeed, research has shown the ACA Medicaid expansions have reduced financial risk for beneficiaries (Hu et al., 2016). The literature pertaining to the relationship between income and risky health behavior is inconclusive (Cawley and Ruhm, 2011); however, it is possible that the reduction in financial risk may induce overeating or greater consumption of unhealthy goods, such as alcohol and cigarettes. At the same time, the income effect could reduce mortality if the financial gains are allocated toward the consumption of healthy goods and behaviors. Thus, the empirical question includes the direction as well as the magnitude of the effects.

Medicaid expansion states and counties differ in level and trend before the expansion, meaning that any direct comparison of expansion with non-expansion groups will be biased. In Figure A.1 we illustrate this issue. The figure depicts estimates from an event-study model using a sample of all counties from 2009–2017, where the outcome variables include the annual, county-level all-cause and amenable mortality rates among adults between 20 and 64 years of age. The figure shows an obvious pre-trend in all-cause and amenable-cause mortality rates in expansion counties relative to non-expansion counties in the years prior to 2014. When we investigated this issue in more detail with specific causes of deaths, the confounding pre-trends are largely explained by differences in cardiovascular and respiratory mortality.

To address these pre-existing differences, we use event study and difference-in-differences models in conjunction with propensity score reweighting. The combination of these techniques provides important advantages relative to either technique in isolation, in both bias and efficiency (Smith and Todd, 2005; Imbens and Rubin, 2015).

⁵ The authors use a model-based approach to estimate the welfare benefit of Medicaid per dollar of government spending. While the model is based on the OHIE, the authors note their estimates are informative for the ACA expansions given the use of national survey data in the model and the similar coverage standard between the OHIE and ACA.

In our setting, the comparison of mortality before and after the reform provides compelling *within*-county variation in treatment status that lends itself to a difference-in-differences approach. What is needed is a control group to address contemporaneous mortality trends in the expansion counties. The selection of a control group is complicated, however, by pre-expansion differences in mortality trends between counties which did or did not expand Medicaid. With this in mind, we use the propensity-score model to select a group of control counties in a way that balances pre-treatment characteristics, including mortality outcomes. Thus, by eliminating control units that are observationally different than treated units, the propensity-score step improves the *across*-county comparison, in terms of both bias and efficiency (Hirano et al., 2003). For use in assessing whether we have created a compelling control group, we exclude four years of pre-period outcomes from the propensity-score model. This exclusion avoids mechanically creating parallel trends in between treatment and control and allows us to interpret the evolution of mortality in this held-out period as a test of quality of the control group. We focus the analysis on the county-level treatment-on-the-treated (county-TOT) effect to answer the question “Did the ACA Medicaid expansion save lives?”⁶

3.1. Propensity-score model

Our first step is to address pre-expansion differences between treatment and control counties by constructing a sample of counties in non-expansion states that is similar to the counties in expansion states. To do so, we fit a propensity-score model that predicts whether a county is located in a state that adopted the Medicaid expansion. The model is estimated using a rich set of demographic, economic, and political characteristics of the states in the years before 2014. Given the large number of potential predictors of treatment and the resulting possibility of overfitting the propensity score, we use the double lasso procedure described in Belloni et al. (2014a) and Urminsky et al. (2016) to select the relevant variables to be included in the model. This method identifies variables for inclusion in the propensity score model in two steps: (1) fitting a lasso regression that predicts the outcome of interest, i.e. the mortality rate for adults ages 20–64, and (2) fitting a lasso regression that predicts the focal independent variable, i.e. expanding Medicaid in 2014.⁷ The union of the

variables estimated to have non-zero coefficients in these steps is then included in the final propensity score model.⁸ As discussed above, we hold out mortality outcomes from the four years preceding the 2014 expansion from the set of potential predictors in the propensity score. This allows us to perform a cross-validation exercise by examining pre-trends in this window. In addition to preventing the overfitting of the propensity score model, the procedure allows for an imperfect selection of controls in the prediction of treatment through the inclusion of predictors of the outcome in the “first-stage” propensity score.

Once we have selected our variables, the propensity score is estimated using the following logistic regression model:

$$\text{Logit}(\Pr(\text{MedicaidExpansion}_c)) = \beta_0 + \beta_1 \mathbf{X}_c, \quad (1)$$

where the dependent variable is an indicator of whether county c expanded Medicaid in 2014. The vector $\mathbf{X}_c$ includes the predictors of expansion selected by the data-driven procedure. Observations with propensity scores that are outside the overlap region are trimmed from the sample. Trimming should serve to increase the internal validity of our estimates at the cost of changing the estimand to apply to the set of matched counties. This cost arises because we cannot infer counterfactual outcomes for counties that do not have an analogue in the counterfactual, although we do investigate alternative trimming thresholds. As pointed out by Imbens and Rubin (2015), excluding unmatched units should improve the asymptotic variance and robustness properties of our estimator. Moreover, the trimmed units tend to be units with high leverage, whose presence makes estimators sensitive to outliers (Young, 2018).

There are several reasons for constructing counterfactuals at the county level as opposed to at the state level. First, the chief advantage relative to a state-level analysis is that we can achieve more precise estimates by modeling sub-state variation. Indeed, visual inspection of state-level trends illustrated in Appendix Figure A.2 reveals little response of mortality rates for adults 20 to 64 years of age in expansion and non-expansion states. Finer geographic data allows us the flexibility to adjust for potentially confounding changes in mortality that would otherwise be unaccounted for when using a broader geographic unit. For instance, suppose that opioid-related mortality is ris-

designed to alleviate these biases by choosing variables that predict both the outcome and/or the focal independent variable.

⁸ We estimate each step using the square-root lasso described in Belloni et al. (2014a) and Belloni et al. (2014b). We allow the selection procedure to choose from a rich set of pre-expansion county-level variables, including the unemployment rate, the poverty rate, logged real median income, logged population, population density, the Obama vote share in 2008, the Obama vote share in 2012, an indicator of whether the state governor was a Democrat in 2010, logged average state health and welfare expenditures between 2005–2013, the pre-expansion uninsured rate for non-elderly adults, the percentage of population in five age groups distributed between 20 and 64 years of age, the percentage that is male, the percentage that is white, the percentage that is black, and the percentage that is Hispanic. We also include the all-cause mortality rates for adults 20 to 64 for each year between 2005 and 2009 as well as the average all-cause, amenable, and non-amenable mortality rates between 2005 and 2013.

⁶ We use the term county-TOT to refer to the treatment effect on county-level mortality estimated in our empirical specifications. This should not be confused with the individual-TOT (ITOT) that can be calculated if given the size of the affected population.

⁷ Besides preventing overfitting, there is an additional advantage of using the double lasso procedure compared to arbitrary variable selection approaches (e.g. Sommers (2017)) or conventional lasso methods. Specifically, excluding variables that have a non-zero relationship with the outcome of interest, either by researcher choice or because these variables are often underestimated using conventional lasso methods, can create substantial regularization and omitted variable bias that adversely affects both estimation and inference (Urminsky et al., 2016). This bias is especially problematic when the excluded variables are important predictors of the focal independent variable. The double lasso procedure is

ing in rural counties at the same time as the Medicaid expansion takes place. By re-weighting the non-expansion counties to have a similar rural share to expansion counties, we can reduce the bias associated with this source of pre-existing difference in mortality trends. County-level data provide richer variation for the re-weighting procedure to account for pre-existing differences in mortality trends. Second, the county level is the smallest geographic unit for which detailed data are available for most variables of interest. The availability of data also enables us to examine heterogeneity by county-level characteristics, such as pre-expansion uninsurance levels. Thus counties maximize the size of potential controls, enabling us to trim outliers, select a closely comparable control group, and retain a large sample. Finally, existing evidence pertaining to the mortality effects of Medicaid has relied largely on county-level analysis. This puts our findings on similar ground to those reported in earlier studies, such as [Sommers \(2017\)](#) and BHNS.

3.2. Event-study and difference-in-differences

After constructing the analysis sample and weights, we run event-study models of the form:

$$Y_{cst} = \alpha_c + \gamma_t + \sum_{\tau=2009, \tau \neq 2013}^{2017} \beta_{\tau} (\text{MedicaidExpansion}_{cs} \times \mathbf{1}[\tau = t]) + \beta \mathbf{X}_{cst} + \epsilon_{cst}, \quad (2)$$

where Y_{cst} indicates deaths per 100,000 people in each observation cell indexed by county c in state s and year t . Included in the model are a full set of county and year fixed effects, indicated by α_c and γ_t , respectively. Time-varying control variables, represented by $\mathbf{X}_{cst}$, are selected using the double lasso procedure for panel data ([Belloni et al., 2016](#)).⁹ These controls reduce standard errors and offer additional controls for any time-varying differences between treatment and control unaccounted for by the reweighting procedure. The coefficients of interest are the β_{τ} 's on the interaction between the indicator for Medicaid expansion and indicator function $\mathbf{1}[\tau = t]$. The event-study model enables us to both test for the presence of pre-trends and capture the evolution of the treatment effect over time.

We then report pooled difference-in-differences results, including models which focus on outcomes for specific age groups, pre-expansion insurance coverage, and cause of death. The pooled model effectively averages the year-specific effects estimates in the event study into a single contrast of pre- and post-expansion differences between

treatment and control counties. The baseline model is specified as follows:

$$Y_{cst} = \alpha_c + \gamma_t + \beta_1 \text{MedicaidExpansion}_{cs} + \beta \mathbf{X}_{cst} + \epsilon_{cst}, \quad (3)$$

where the coefficient on *MedicaidExpansion* represents the change in mortality (per 100,000 people) among counties in states that expanded Medicaid in 2014 relative to counties in states that did not. Our preferred specification includes the time-varying controls selected by the double lasso procedure. Throughout the analysis we cluster standard errors by state to account for the fact that our variation comes from state-level adoption decisions. We weight our regressions by the county population 20 to 64 years of age multiplied by $T + (1 - T) \times \frac{p}{(1-p)}$, where T is an indicator for treatment and p is the estimated propensity score. Weighting in this way provides a consistent estimator of the average county-TOT ([Imbens, 2004](#)).

The empirical strategy and data we use differ from the models evaluated in the power calculations in BHNS in three key ways. First, we focus on a fully-specified, data-driven propensity-score model, which has the advantages in both bias and efficiency we have discussed above.¹⁰ In contrast, BHNS evaluates models with evident pre-trends, which they acknowledge significantly reduces power. Second, we use two additional years of data relative to the power calculations in the most recent version of BHNS. Given the growing effects, this increases the likelihood of detecting an effect both due to an increase in the precision and likely magnitude of the effects. Finally, we report reduced-form estimates for uninsurance and mortality while remaining agnostic as to our ability to estimate precise treatment-on-the-treated estimates. In contrast, BHNS draw their strongest conclusions (especially in the introduction of their paper) when using a first-stage estimate that is likely considerably smaller than the group of individuals actually affected by the reform. We discuss the magnitude and interpretation of our estimates at the conclusion of Section 4.

To summarize, we do not believe the power calculations in BHNS are informative about the estimates derived from our model.¹¹ We provide permutation tests to assess our design in Section 5.

⁹ We allow the selection procedure to choose from time-varying demographic and economic controls. These include population percentages in five age groups (20–24, 25–34, 35–44, 45–54, and 55–64), three racial groups (white, black, and other), two sex groups (male and female), and the percentage of population that is Hispanic. We also include logged total county population, the logged population among adults between 20 and 64 years of age, and logged population variables for each age, racial, and sex group. Finally, we include county poverty rates, unemployment rates, and median household incomes. Selected controls are indicated in the table notes.

¹⁰ Although BHNS reports empirical estimates from a propensity score model, the authors do not perform their power calculation exercise for this model. Additionally, the empirical procedure in their propensity score model differs substantially from ours by: excluding a number of important predictors, such as political variables, from the propensity score; declining to use any variable selection procedure, thereby leading to the inclusion of spurious predictors in the propensity score; and, declining to assess overlap or trim on the propensity score. These differences in the estimation are likely to increase bias and variance relative to the model here.

¹¹ There are other notable differences between the papers. For example, BHNS use a large set of time-varying controls in almost all specifications, which likely absorbs some effect of the reform. Instead, we use the panel-lasso model to select an appropriate set of parsimonious controls in those specification in which we include them. BHNS also switch between a 5-year pre-period in their difference-in-differences models and a 10-year pre-period in the event study. Geographic mortality trends have evolved a great deal since 1999, likely leading to imprecision in, and possibly bias from, the estimates of the county effects in a longer panel. Empirically, pre-trend performance is poor in these early years, especially in their triple-difference model. When we re-run our model with a 10-year pre-

Table 1
Summary statistics.

Variables	Expansion counties Mean (SD)	Matched counties Mean (SD)
<i>Baseline mortality rates</i>		
All cause mortality (per 100,000 people)	315.17 (94.69)	359.06 (117.06)
Males	394.31 (120.53)	449.76 (149.54)
Females	237.43 (75.79)	271.23 (94.93)
Ages 20–24	82.03 (46.25)	95.89 (62.41)
Ages 25–34	96.64 (46.99)	114.01 (54.57)
Ages 35–44	160.49 (65.42)	184.84 (80.28)
Ages 45–54	379.59 (119.13)	434.35 (147.28)
Ages 55–64	795.59 (196.38)	885.72 (236.04)
Amenable cause mortality (per 100,000 people)	200.78 (61.56)	226.80 (78.03)
Males	239.45 (76.33)	272.11 (98.36)
Females	162.88 (52.74)	183.08 (66.75)
Ages 20–24	11.69 (13.35)	13.18 (19.27)
Ages 25–34	23.83 (15.18)	27.84 (20.07)
Ages 35–44	71.68 (31.64)	83.48 (42.64)
Ages 45–54	238.82 (80.31)	273.46 (100.40)
Ages 55–64	607.86 (158.58)	668.63 (189.14)
Cardiovascular mortality (per 100,000 people)	63.25 (25.32)	72.03 (32.06)
Respiratory mortality (per 100,000 people)	17.47 (10.24)	21.17 (13.44)
Cancer mortality (per 100,000 people)	90.81 (24.30)	98.58 (31.64)
Suicides (per 100,000 people)	15.28 (7.45)	18.19 (8.77)
Opioid overdoses (per 100,000 people)	2.66 (3.89)	3.67 (4.30)
Drug & alcohol poisonings (per 100,000 people)	13.23 (9.26)	12.95 (9.12)
<i>Independent variables</i>		
% Population ages 20–24	0.12 (0.03)	0.12 (0.04)
% Population ages 25–34	0.23 (0.03)	0.22 (0.03)
% Population ages 35–44	0.22 (0.02)	0.22 (0.02)
% Population ages 45–54	0.24 (0.02)	0.24 (0.02)
% Population ages 55–64	0.20 (0.03)	0.20 (0.04)
% Male	0.50 (0.01)	0.49 (0.02)
% Hispanic	0.19 (0.16)	0.16 (0.19)
% White	0.78 (0.14)	0.79 (0.13)
% Black	0.11 (0.11)	0.15 (0.12)
Unemployed rate	0.09 (0.02)	0.08 (0.02)
Poverty rate	0.15 (0.05)	0.16 (0.05)
Real median income (\$10,000)	5.95 (1.46)	5.33 (1.49)
Uninsured rate	0.19 (0.06)	0.24 (0.07)
Obama 2008 vote share	0.46 (0.12)	0.38 (0.13)
Obama 2012 vote share	0.43 (0.13)	0.35 (0.14)
Democratic governor	0.76 (0.43)	0.49 (0.50)

Notes: The above table presents population weighted means and standard deviations of baseline variables, measured for 2009–2013, in counties within states that expanded Medicaid in 2014 and counties in non-expansion states.

3.3. Propensity score estimates

We first report predictors of expansion selected by the double lasso procedure. The procedure selects county age, race, economic measures, political variables, and pre-expansion mortality rates as predictors of expansion. Counties with large population percentages in the 45-to-54 and 55-to-64 age groups are more likely to implement the expansion. For males and Hispanics, higher population shares are positive predictors of expansion, while higher shares of blacks are negative predictors. Expansion counties had higher median incomes, but also higher unemployment rates. The pre-expansion non-elderly uninsured rate negatively predicts expansion. Having Democratic governors at the time of the passage of

the ACA and having high 2008 and 2012 Obama county vote shares are both strong predictors. Finally, having higher non-elderly mortality rates in the years prior to the ACA is a modest predictor of expansion, with small and mainly insignificant estimates across these years. The full set of coefficient estimates appear in Appendix Table A.2.

The resulting propensity-score distributions of the 2823 treatment and control counties display significant overlap for nearly the entire distribution, except that there are few non-expansion counties in the very upper tail. We trim the sample at areas in the tails where there is no overlap (propensity scores outside of 3.8 and 97.1 percent). Trimming removes 563 counties from the sample, 78 in expansion states and 485 in non-expansion states. After trimming, our final sample consists of 2260 counties, 1174 and 1086 in expansion and non-expansion states, respectively. The distribution of the propensity-score and a map of counties appear in Appendix Figure A.3 and Appendix Figure A.4, respectively. Summary statistics are presented in

period, we find standard errors are more than 50 percent larger than in our reported estimates.

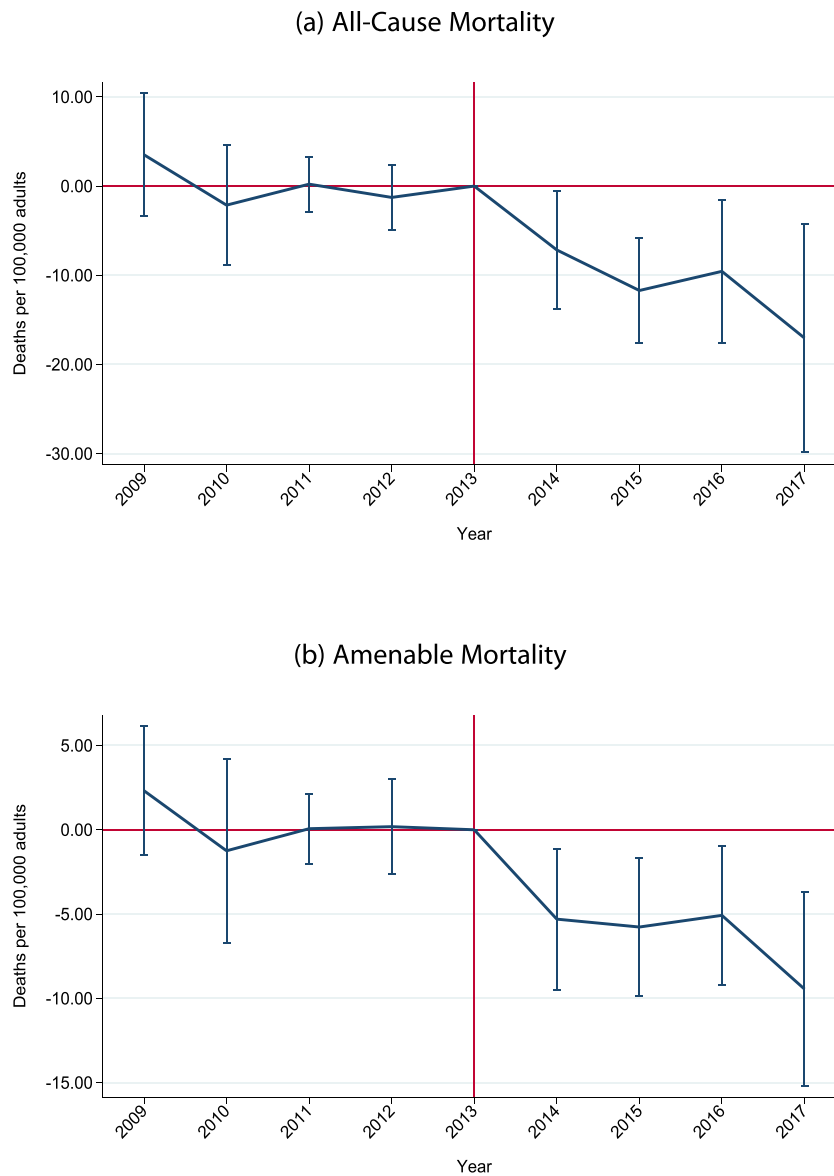


Fig. 1. Event study: ACA medicaid expansion and mortality. *Notes:* The above figure shows event-study plots of Medicaid expansion on (a) all-cause mortality and (b) amenable causes of mortality for adults aged 20 to 64 year. The vertical line indicates the year prior to the effective date of when expansion. Bands indicate 95 percent confidence intervals. Standard errors are clustered at the state level. Models include year and county fixed effects. Controls include the county unemployment rate, the percentage of population that is white, the percentage of population that is between 55 and 64 years of age, the logged county population of adults between 20 and 64 years of age, the logged county population of adults between 35 and 44 years of age, and the logged county population of females between 20 and 64 years of age.

Table 1. Treatment counties generally have slightly lower baseline mortality rates. Treatment and control counties are similar among the independent variables except that treatment counties are more likely to be located in a state with a Democratic governor in 2010. For comparison, we also present summary statistics for the trimmed counties in Appendix Table A.3. The trimmed counties differ greatly from the counties that remain in the sample, particularly in terms of the baseline mortality rates, income, uninsured rates, and the political variables included in the propensity-score model.

4. Main results

4.1. Medicaid and mortality

We begin by assessing pre-trends in mortality outcomes, as reported in Fig. 1. In Fig. 1(a), point estimates for all-cause mortality are near zero and flat through 2013. Recall that year-by-year mortality outcomes in 2010 to 2013 are not included in the propensity score model, so the flat pre-trends in this time frame serve as a crucial check on the ability of the propensity score model to construct

Table 2
Effect of ACA medicaid expansion on mortality.

Model and variable	Full sample		Gender		Age groups					Uninsured	
	Base	Controls	Males	Females	20–24	25–34	35–44	45–54	55–64	High	Low
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
<i>Panel A: All cause mortality</i>											
Medicaid expansion	–14.83** (6.12)	–11.36*** (3.59)	–11.36*** (4.27)	–11.60*** (3.31)	–0.95 (2.61)	–5.19 (3.99)	–7.93** (3.42)	–15.35* (8.51)	–20.65* (11.09)	–12.40*** (4.22)	–3.96 (4.42)
% Effect relative to baseline	–4.71	–3.60	–2.88	–4.89	–1.16	–5.37	–4.94	–4.04	–2.60	–3.81	–1.30
<i>Panel B: Amenable cause mortality</i>											
Medicaid expansion	–7.79*** (2.15)	–6.64*** (1.93)	–5.94*** (2.13)	–7.41*** (2.15)	0.29 (0.39)	1.07 (1.21)	–2.44* (1.37)	–8.12* (4.86)	–20.22*** (6.55)	–7.89*** (2.69)	–2.88* (1.49)
% Effect relative to baseline	–3.88	–3.31	–2.48	–4.55	2.48	4.49	–3.40	–3.40	–3.33	–3.84	–1.74
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clusters	45	45	45	45	45	45	45	45	45	40	38
Observations	20340	20340	20340	20340	20340	20340	20340	20340	20340	18882	11214

Notes: Observations include annual county-level mortality rates for adults aged 20 to 64 years. Columns (1) and (2) include results for the full sample with and without controls. In columns (3) and (4) we report results for males and females, respectively. In columns (5) through (9) we report results for five separate age groups. In columns (10) and (11) we report heterogeneous effects by county baseline uninsured levels, where *High/Low* uninsured counties are defined as being above/below the median baseline uninsured rate for individuals aged 19 to 64 years. Coefficients indicate deaths per 100,000 people. Standard errors are clustered at the state level. All models include year and county fixed effects. Controls include the county unemployment rate, the percentage of population that is white, the percentage of population that is between 55 and 64 years of age, the logged county population of adults between 20 and 64 years of age, the logged county population of adults between 35 and 44 years of age, and the logged county population of females between 20 and 64 years of age.

balanced treatment and control groups. When we disaggregate all-cause mortality to examine amenable causes in Fig. 1(b), we again find flat pre-trends. We conclude from these figures that the propensity-score model selects a comparison group with similar pre-expansion mortality patterns. Given that the ACA was passed in 2010, and some changes went into effect before 2014, the patterns indicate that any effects before 2014 were uncorrelated with the later Medicaid expansion.

Moving to the post-expansion period, we see a notable reduction in all-cause mortality within the first four years of the expansion. Mortality falls in the first year after expansion and is lower by approximately 10 deaths per 100,000 inhabitants by 2015 and 2016. In 2017 we find an even larger reduction in mortality of over 15 deaths per 100,000 inhabitants. The timing of the effect aligns with the expansion and shows mortality reductions became larger over time. Disaggregating the post-expansion effects by causes of death that may be amenable to healthcare reveals that amenable causes account for the majority of the effects in aggregate mortality.

Table 2 reports difference-in-differences estimates for the full sample, by gender and age group, and by pre-expansion uninsured levels. These results effectively pool together the 2014 to 2017 estimates reported in the event study. In column 1, we find that all-cause mortality drops by 14.83 deaths per 100,000 inhabitants. Adding controls in column 2 only slightly reduces the point estimate to 11.36 deaths per 100,000 averted, a 3.60 percent decrease in mortality. The reduction in effect size may occur due to time-varying differences between treatment and control captured by the controls; however, we cannot rule out that the difference reflects statistical noise. Both estimates are statistically significant at or below the conventional 5 per-

cent level. We proceed with the model with controls as our preferred specification.

In columns 1 and 2 of Panel B, we see that the majority of the aggregate mortality effect is accounted for by changes in amenable causes of death, i.e. those which may be averted by access to optimal healthcare. The estimated reduction in amenable cause mortality is 7.79 and 6.64 deaths per 100,000 people with and without controls, respectively. Confidence intervals around these estimates rule out no response. The effects on amenable cause mortality are the strongest evidence of mortality reduction following the reform. Non-amenable causes of death make slightly smaller and statistically insignificant contributions to the total and are substantially less robust to the inclusion of controls compared to amenable causes of death (coefficient = -7.04 se = 4.45 without controls, coefficient = -4.72 se = 2.54 with controls). We tested whether the coefficients for amenable and non-amenable mortality are statistically different from one another. The results indicate that we cannot reject the null hypothesis that the coefficients are the same (p -value = 0.48). There are reasons to interpret evidence on non-amenable mortality with caution—due to the difficulty in classifying deaths, non-amenable causes of death may also respond to the Medicaid expansions. For instance, Doyle Jr. (2005) and Howell et al. (2010) show health insurance is associated with reductions mortality after automobile accidents and external injuries among individuals below 20 years of age, respectively.

In the middle columns of Table 2 we report results by gender and age group.¹² In columns 3 and 4, the

¹² We caution against drawing strong conclusions from individual sub-analyses for several reasons. First, our research design and focal hypothesis test intends to estimate all-cause population mortality effects,

reported results imply that mortality effects are nearly equal between men and women in levels, but smaller in percentage terms for men, given their higher baseline mortality rates. As with the combined effect, we find larger and more precisely estimated reductions in amenable causes of death for each gender, which can explain the estimated effects on all-cause mortality for both genders. We also find flat pre-trends for male and female mortality rates, as shown in Appendix Figures A.5 and A.6. We cannot reject equality between the male and female mortality reductions for all-cause (p -value = 0.95) and amenable deaths (p -value = 0.45). The analysis of effects by age groups finds the largest effects for individuals approaching Medicare eligibility, in the 55-to-64 age group. Importantly, we find no evidence of differences in all-cause or amenable mortality trends for the near elderly and the 45-to-54 age group prior to the expansions taking effect. We do, however, find differential pre-trends for adults 20-to-24, 25-to-34, and 35-to-44 years of age. Although the younger adults contribute a modest amount the total all-cause and amenable mortality effects we estimate, we nevertheless refrain from making a strong causal interpretation of the estimates for the adults under 45 years of age. Comparing estimates across age groups, we can reject equality of all-cause mortality reductions for the 55-to-64 age group and adults 20 to 24 years of age (p -value = 0.02). For amenable causes, we can reject equality of mortality reductions for the 55-to-64 age group and adults in age groups below 45 years of age (p -values = 0.00). We cannot, however, reject equality in all-cause and amenable mortality impacts across other age groups.

In columns 10 and 11 we report results in counties with high and low rates of uninsured individuals. To derive these results, we first calculate the median uninsured rate for non-elderly adults in expansion counties between the years 2005 and 2013. We then separate expansion counties by whether they are above or below the median rate, 17.04 percent, and match the counties in both groups to the pool of all non-expansion counties using variables selected by double lasso.¹³ We then estimate mortality effects for both samples by reweighting our regressions with the estimated propensity scores multiplied by county populations. Our estimates imply a reduction of 12.40 deaths per 100,000 residents in expansion counties above the median baseline uninsured rate. This estimate is largely accounted for by a drop in amenable cause mortality. For expansion counties with relatively low rates of uninsured individuals, the all-cause and amenable mortality impacts are small and neither is significant at the 5 percent level. Event-study figures illustrated in Appendix Figure A.7 display relatively-flat pre-trends and sharp breaks in the all-cause and amenable mortality trends for high uninsured counties, consistent with the pattern of the main estimates.

as opposed to sub-groups. Second, there is potential for multiple hypothesis testing issues as we disaggregate the analysis.

¹³ An earlier version of this paper split the sample by the full-sample uninsurance rate, resulting in 69 percent of treatment counties assigned to the low uninsurance group. Results from this specification also showed larger mortality losses in high-uninsured counties, though the gap was smaller.

The figures for low uninsured counties, however, exhibit more noisy pre-trends and no clear response in the post-expansion period.

We report results for the detailed cause-of-death analysis in Appendix Table A.5. We include common causes of death among adults 20 to 64 years of age, as indicated by Appendix Figure A.8. For many of the groups we analyze, namely younger adults, mortality rates for specific causes of death are very low and we may not be adequately powered to estimate the mortality responses. We find reductions in cardiovascular and respiratory-related mortality (e.g. due to heart attacks, strokes, chronic obstructive pulmonary disease, asthma). We do not find evidence that cancer-related deaths have responded to the expansions, although the estimate for the near-elderly is narrowly insignificant. Event-study plots, illustrated in Appendix Figures A.9, A.10, and A.11, show flat pre-trends for models that aggregate ages 20 to 64 together as well as for the specific age-groups that explain most of the aggregate mortality reductions, namely adults 45 to 54 years of age for cardiovascular diseases and 55 to 64 years of age for respiratory- and cancer-related deaths. These results echo previous research that has shown fewer uninsured cardiac surgery patients and improved predicted risk scores and morbidity rates among cardiovascular related illnesses in states that expanded Medicaid (Charles et al., 2017) as well as substantial increases in the use of cardiovascular and respiratory related prescription medications (Ghosh et al., 2017). Most people with cardiovascular or respiratory diseases experience very large benefits from access to the correct medication, such as statins for cardiovascular disease. There is persuasive evidence that connects higher Medicaid prescription drug spending reduces mortality, particularly among cardiovascular, circulatory, and respiratory diseases (Clayton, 2019). Furthermore, the medical literature finds these medications work quickly, consistent with the immediate drop in mortality we find. For example, statins used to treat or prevent cardiovascular diseases generally take 4 to 6 weeks to stabilize a patient's cholesterol levels (National Health Service, 2018). Thus, responses on these disease margins are consistent with the potential for large gains from effective treatments that could be accessed through Medicaid (Centers for Disease Control and Prevention, 2011).

We also examine the effect of Medicaid expansion on diseases of despair, including suicide, opioid overdoses, and drug and alcohol poisonings.¹⁴ These causes of death, particularly opioid overdoses, have risen over the previous two decades. While the connection between these causes of death and Medicaid is more complex than with cardiovascular diseases and respiratory disease, access to healthcare may also influence them. For suicide, the results suggest reductions in the 25-to-34 and 55-to-64 age groups, although the event-study figure for the latter age group displays differential pre-trends. The reduction in suicide is consistent with evidence from Maclean et al. (2017), who show that Medicaid expansion states experienced signifi-

¹⁴ Event-study plots for these causes of deaths appear in Appendix Figures A.12, A.13, and A.14.

cant increases in mental health services and psychotropic prescriptions. Opioid overdoses and drug and alcohol poisonings show little response. Given the concerns about multiple testing and the imprecision of the estimates, we conclude that there is insufficient statistical power to evaluate the effect of the Medicaid expansion on these causes of death.

4.2. Interpretation of results and relation to other estimates in literature

By any measure, these point estimates imply meaningful reductions in mortality have resulted from the ACA Medicaid expansion. Comparisons of the magnitudes to other estimates in the literature depends, however, on the definition of the affected, or “treated,” population. To place the estimates in context, we begin by relating them to the measured change in uninsurance before considering broader groups of affected individuals in this and other studies.

If we assume that all effects are concentrated among individuals who switch from uninsurance to Medicaid, ITOT effects can be calculated based on a 4.15 percent gain in health insurance among all non-elderly adults we estimate using county-level health insurance as the outcome for the same sample, specification, and weights as the primary mortality regression.¹⁵ Similarly, we can calculate ITOT effects for adults 55 to 64 years of age, the age group that largely explains the aggregate mortality reductions, based on a 2.77 percent gain in insurance.¹⁶ Calculating the ITOT additionally requires an adjustment for the higher mortality of our estimates by the ratio of affected to average mortality, as many studies find evidence that individuals in worse health are more likely to enroll in Medicaid (e.g. Kenney et al. (2012), Marton and Yelowitz (2015), Miller et al. (2019)). Specifically, we combine the first-stage estimates with national mortality data that includes mean individual incomes, deaths, and population counts from Chetty et al. (2016). Adjusting for the higher mortality rates among the Medicaid population relative to average, the estimated ITOT effects imply a 30 percent reduction in mortality risk for adults aged 20 to 64 years as well as the near-elderly population. Taken together, these calculations imply that one death was averted for every 310 newly-covered individuals 20 to 64 years of age and 180 newly-covered individuals aged 55 to 64 years.¹⁷

¹⁵ Results for the full population and those with incomes below 138 percent of the poverty level appear in Appendix Table A.4. The results are consistent with the findings reported in Miller and Wherry (2019); see Fig. 1, panel a.

¹⁶ SAHIE does not provide data for each age group we analyze but it does provide data for adults 50 to 64 years of age. While not perfect, this data closely overlaps the near-elderly population we study. Survey datasets, such as the National Health Interview Survey, provide an alternative source of insurance information by age. Research shows, however, that insurance enrollment data from survey data suffers from under-reporting Miller et al., 2019.

¹⁷ The Chetty et al. (2016) data is limited to ages 40 and older, so the adjustment uses ages 40 to 64 in 2013. Without the mortality scaling, our estimated effect size combined with the estimated first-stage implies an 86 percent reduction. The mean mortality rate in this age range is

While these ITOT calculations provide an informative scaling of the results, we prefer to interpret the results as reduced-form estimates of the policy. The concern is that the ITOT scaling hinges on the size of the affected population, which may extend significantly beyond the measurable change in uninsurance in standard datasets.¹⁸ There are numerous ways in which the ACA Medicaid expansion plausibly affects a larger population than is reflected by the net change in uninsurance. First, federal law mandates coverage of medical bills up to three months before the application to join Medicaid. This exposes virtually all low-income individuals to treatment in some form. Second, the ACA gives hospitals the ability to grant “presumptive eligibility,” temporary Medicaid enrollment, to patients who have incomes that are low enough to qualify for the program. Thus, in the most critical life-saving situations, the ACA Medicaid expansion guarantees reimbursement for care provided to nearly all targeted individuals. Third, individuals may cycle off and on Medicaid. This type of churn has been documented previously and would mean that a snapshot measure of coverage will understate the affected group. Fourth, Medicaid induces higher rates of healthcare use through lower levels of cost-sharing compared to private insurance plans. Several studies find that larger amounts of cost-sharing, a common feature for bottom-of-the-market insurance plans, is associated with less healthcare spending and worse health outcomes, particularly for low-income individuals (Manning et al., 1987; Haviland et al., 2011; Wharam et al., 2013). Therefore, it is plausible that low-income individuals who switched from private insurance plans to Medicaid experienced health gains. Fifth, there have been documented effects on financial well-being, crime, and quality of care, all of which likely creates spillovers onto ineligible individuals. For all of these reasons, we interpret the evidence as consistent with effects accruing in a broader population than those who receive insurance as a result of Medicaid expansion.

approximately 344 deaths per 100,000 people. We then calculate the average mortality rate for individuals with incomes at or below 138 percent of the federal poverty level. This corresponds to about 1037 deaths per 100,000 using the income threshold for individuals and 967 deaths per 100,000 using the threshold for a household of two. The adjusted mortality effect for adults 20 to 64 is then between $0.86 \times \frac{344}{1037} = 0.29$ and $0.86 \times \frac{344}{967} = 0.31$. For the per beneficiary effects, a 4.15 percent first stage implies 4,436,350 newly covered individuals based off a population in expansion states of approximately 106,900,000. Using the death rates for the total (344) and expansion (1037) population, a 3.60 percent reduction in mortality corresponds to 14,345 reduced deaths, and $\frac{4,436,350}{14,345} = 310$. For individuals 55 to 64 years of age, our estimated effect size combined with the estimated first-stage implies a 90 percent reduction in mortality without adjusting for differences in mortality rates among those who are Medicaid eligible. The death rate for individuals with incomes at or below 138 percent of the federal poverty level is 1700 per 100,000 people. The adjusted mortality effect is then $0.90 \times \frac{564}{1700} = 0.30$. For the per beneficiary effect, a 2.77 percent first stage implies 592,226 newly covered individuals based off a near-elderly population of approximately 21,380,000. Using the death rates for the total (564) and expansion (1699) population, a 2.60 percent reduction in mortality corresponds to 3309 reduced deaths, and $\frac{592,226}{3309} = 180$.

¹⁸ Our discussion borrows from Miller et al. (2019), who provide a detailed discussion of the policy environment and issues with survey-based measures of uninsurance.

Despite the uncertainty in the ITOT scaling, our estimated treatment effects fall within the range of most estimates derived from previous studies that investigate the relationship between Medicaid and mortality. Among the pre-ACA Medicaid studies, our paper most closely aligns with [Sommers \(2017\)](#), who studies mortality effects following expansions in Arizona, Maine, and New York in the early 2000s. The TOT effect presented by Sommers is about twice as large compared to our preferred county-TOT estimate (-19.1 vs. -11.36 deaths per 100,000 people).¹⁹ Our estimated reduction in individual mortality risk of 30 percent falls, however, well within the range estimated by Sommers (between -38 and -25 percent).²⁰ Our findings are also comparable to the pre-ACA mortality estimates produced by the OHIE. The authors find a 0.0013 percentage point decrease in the probability of death compared with the control group. This estimate, however, is very imprecise. The 95 percent confidence interval includes effect sizes between -0.0066 and $+0.0040$. Based off a baseline mortality rate of 0.008, their estimates imply an individual-level mortality change of -16.3 percent, with confidence intervals that include a -82 and $+50$ percent change. The -30 percent ITOT effect we find is approximately twice as large as the effect derived from the OHIE. This comparison is imperfect, however, given that the OHIE first-stage estimate corresponds to the change in Medicaid coverage, as opposed to the total change in insurance coverage among Medicaid and non-Medicaid eligible individuals. If we use the 8.18 percent change in Medicaid coverage as an alternative first stage, our ITOT estimate is reduced to -17 percent.

Similarly, our work can be compared to more recent work that has examined the mortality impacts of the ACA expansion. First, [Swaminathan et al. \(2018\)](#) examine the impact of the ACA Medicaid expansions on the one-year survival rate of patients with end stage renal disease. Individuals in this population have a substantially higher risk of mortality relative to the average adult. The researchers report that expansions lead to a 8.5 percent reduction in mortality for these individuals, an effect size much larger than the percentage reduction in all-cause mortality we find. Second, our results differ from the empirical estimates in BHNS, who find a statistically insignificant reduction of 3 deaths per 100,000 using a triple-difference strategy that estimates changes in mortality among the near-elderly

population using a separate control group of adults ages 65 to 74 years. In addition to the triple-difference approach, BHNS also report similar results using a propensity-score reweighting design. As discussed in section 3.1, methodological choices likely explain an important portion of the differences in our findings. In addition, our data includes an extra year of mortality outcomes as compared to BHNS. As our year-by-year point estimates are growing over time, an extra year of outcome data contributes precision for reasons beyond the increase in sample size.

Two papers released since our first working paper have provided additional evidence on the effect the ACA Medicaid expansion on mortality reductions in low-income populations. Both papers possess strong claims to internal validity. First, [Miller et al. \(2019\)](#) examines the mortality of ACS respondents through a link to administrative Census data. Their estimates imply ITOT effects of 23 and 64 percent reductions in mortality for adults 20 to 64 and 55 to 64 years of age, respectively. The authors also advocate for a similar reduced-form interpretation of the results in light of the reasons we discuss above. Second, a recent working paper by [Goldin et al. \(2019\)](#) evaluates an intervention where in 2017, the Internal Revenue Service sent informational letters to nearly 4 million randomly chosen taxpayers who had previously paid an income tax penalty for lacking health insurance coverage under the individual mandate provision of the ACA. The purpose of the letters encouraged take-up of health insurance in some form, either through the individual marketplace or public insurance programs like Medicaid. The authors find the intervention increased insurance take-up among individuals who were uninsured for some portion of the year prior to receiving a letter by 1.3 percentage points in the year following the intervention, a 2.8 percent increase compared to individuals that met the criteria to receive a letter but did not. The coverage effects are largest among individuals who lacked any coverage and among older, non-elderly adults. Using the treatment group assignment as an instrument for coverage, the authors estimate that the average per-month effect of the coverage induced by the intervention on two-year mortality was approximately -0.17 percentage points for adults 45 to 64 years of age, an 11.9 percent reduction. Consistent with our findings, the authors find no evidence that the intervention reduced mortality among younger adults. Comparing our results to this study is challenging because we focus on the relationship between mortality and Medicaid, not mortality and insurance more generally. If we assume, however, the mortality estimates reported in Goldin, Lurie, and McCubbin do apply to those covered by Medicaid, we can scale their estimates to make a more direct comparison. [Gordon et al. \(2019\)](#) finds the typical duration of Medicaid coverage lasts for approximately one year, so multiplying the reported per-month estimate of -0.17 by 12 yields an effect of annual coverage on mortality of -2 percentage points.²¹ If we also double the reported

¹⁹ To make an apples-to-apples comparison and replicate, with modifications, the analysis reported in [Sommers \(2017\)](#), we re-estimate our model using the early-2000s Medicaid expansions in Arizona, New York, and Maine. Details of this analysis appear in Appendix B. We generate short- and long-run estimates by restricting the follow-up period to the first four and eight years following these earlier reforms, and estimate effects on aggregate and amenable mortality. Although mortality falls following the reforms (our point estimates are approximately one-half the size of the prevailing estimates of the effects of Medicaid on mortality), it is difficult to distinguish these improvements in mortality from a trend extrapolated from before the reforms, particularly for amenable causes of death. Evidence of pre-trends suggests that we cannot draw strong causal conclusions from these reforms using our methodology.

²⁰ To calculate the per-beneficiary estimates, our approach (described above) differs from that of Sommers in that he uses a per-beneficiary calculation is derived using a model that interacting the indicator for whether a state expanded Medicaid with the endogenous change in insurance coverage.

²¹ We note that this scaling is imperfect because [Goldin et al. \(2019\)](#) find evidence of a concave relationship between coverage and mortality, so multiplying the reported per-month estimate by 12 likely over-estimates the effect of annual coverage on mortality.

Table 3

Alternative specifications: effect of Medicaid expansion on mortality.

Model	All cause		Amenable	
	(1)	(2)	(3)	(4)
Preferred estimates	-14.83** (6.12)	-11.36*** (3.59)	-7.79*** (2.15)	-6.64*** (1.93)
No pre-treatment outcomes	-12.97** (6.05)	-10.12*** (3.58)	-7.10*** (1.99)	-6.35*** (1.95)
Imbens selection procedure	-14.52** (6.27)	-11.73*** (3.75)	-7.22*** (2.08)	-6.27*** (2.03)
Ridge regression	-10.76* (6.04)	-10.55*** (3.22)	-6.42*** (2.07)	-6.61*** (1.81)
1:1 nearest neighbor matching	-14.98*** (5.17)	-9.37** (3.71)	-7.22*** (1.83)	-5.48** (2.27)
2:1 nearest neighbor matching	-15.98*** (5.59)	-11.54*** (3.56)	-8.05*** (2.32)	-6.36*** (2.04)
Adding late expanding states	-11.22** (5.57)	-10.10*** (3.16)	-6.77*** (1.95)	-6.60*** (1.55)
Exclude early expanding states	-12.74** (6.43)	-9.13** (3.82)	-6.70*** (2.23)	-5.48*** (1.80)
Average treatment effect	-17.15*** (5.91)	-14.36*** (3.15)	-8.80*** (2.27)	-7.60*** (1.93)
County-specific trends	-14.75** (6.46)	-12.06*** (3.91)	-7.74*** (2.27)	-6.96*** (2.06)
Logged deaths	-0.031*** (0.009)	-0.030*** (0.009)	-0.026*** (0.009)	-0.028*** (0.009)
Trim propensity score.1/.9	-6.66 (5.68)	-7.42* (4.16)	-5.96*** (2.17)	-5.52** (2.15)
Controls	No	Yes	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes
County fixed effects	Yes	Yes	Yes	Yes

Notes: Observations include annual county-level mortality rates for adults ages 20 to 64 for the years 2009–2017. Coefficients indicate deaths per 100,000 people. Standard errors are clustered at the state level. All models include year and county fixed effects. Controls selected using double lasso. In the first row, we show estimated mortality effects derived using our preferred specification. In the second row, we show estimated mortality effects without including pre-treatment outcome variables in the matching procedure. In the third row, we present estimated mortality effects using the Imbens variable selection procedure. In the fourth row, we present estimates where we use ridge regression to select variables for the propensity score. In rows five and six we present estimate using 1:1 and 2:1 nearest-neighbor matching, respectively. In row seven we include states that expanded Medicaid after 2014 (AK, IN, NH, PA and MT). In row eight we excluded states that expanded Medicaid prior to 2014 (DC, DE, MA, NY and VT). In row nine we present estimates of the average treatment effect. In row 10 we include county-specific linear time trends to the regressions. In row 11 we define the outcome variables as logged deaths (plus one). In the final row, we trim the sample of counties with propensity scores below 0.1 or above 0.9.

two-year baseline mortality rate of 1.4 percent to match our four-year follow-up (2.8 percent), the -2 percentage point reduction in two-year mortality implies an annual coverage reduction in four-year mortality of 71 percent, with confidence intervals that include our estimated 30 percent reduction in individual mortality risk.

5. Robustness

In this section, we first discuss results from a wide range of alternative specifications. We then present and discuss results from a permutation test designed to gauge the likelihood that our model is indeed powered to estimate causal mortality effects. We conclude this section by discussing mortality estimates using elderly adults as an additional control group.

5.1. Alternative specifications

We conduct a substantial range of robustness analyses, reported in Table 3. In the first row, we report results derived from our preferred specification. In the next five rows we use a number of alternative methods for the propensity score adjustment. First, we report results where

we exclude pre-treatment outcomes in the double-lasso procedure. We then report results derived from implementing the data-driven method outlined in Imbens and Rubin (2015) to select variables for the propensity-score model. Next, we show estimates where we replace the double-lasso method with ridge regression for model selection. Finally, we present estimates using 1-to-1 and 2-to-1 nearest-neighbor matching (with replacement) in place of reweighing. We implement double lasso to select variables for these matching models. In all alternative specifications for model-selection, we find results that are very similar to our preferred estimates in both magnitude and precision.

We next consider robustness to the definition of treatment that focuses on initial adopters of the 2014 expansion. To do so, we add back the few states (PA, IN, NH, AK, MT, LA) that implemented the Medicaid expansion after the first half of 2014. We then exclude five states that are included in our main analysis (DC, DE, MA, NY, and VT) but chose to expand Medicaid or similar coverage to low-income adults from 2009 through 2013 (Miller and Wherry, 2017). The results from these specifications do not differ from our preferred estimates. Finally, we move from estimating the county-TOT to estimating average treatment effects (county-ATE). The county-TOT estimates we

present apply to the group of counties that were treated. The county-ATE effects should be interpreted as effects among the population that, in principle, *could have* been treated. This includes both expansion and non-expansion counties. Here we weight our regression by population multiplied by $\frac{p}{1-p}$, where p is the estimated propensity score. The average treatment effects show moderately larger effect sizes, suggesting that non-expansion states may have benefited more from the Medicaid expansion than those that adopted. This finding is consistent with the high-low uninsurance findings reported above. Counties in non-expansion states had higher baseline uninsured rates in addition to other characteristics that would be expected to correlate with higher rates of uninsurance.

We also include county-specific linear trends to address concerns pertaining to county-level unobserved characteristics that may be evolving over time and correlated with mortality (e.g. changes in population compositions between counties with different ex ante insurance rates), potentially biasing our results. Including the time trends does little to change the estimates, reinforcing evidence that our model is not confounded by unobserved trends between expansion and non-expansion counties. We then estimate our model using logged deaths as opposed to death rates. We first add one death to each observation to avoid excluding (the very small number of) counties that experienced no deaths. The results using logged deaths closely mirror our estimates using mortality rates as the outcome, with estimates suggesting a 3 percent reduction in all-cause and amenable mortality. Finally, we explore how estimates change when we use an alternative trimming of our sample. This exercise also serves to remove units that tend to carry high leverage and make estimators sensitive to outliers. While the trim helps discipline our estimates, it comes at the cost of changing the estimand to reflect causal effects for the matched (i.e. not trimmed) counties. We trim counties with propensity scores that are less than 0.1 or greater than 0.9, the extreme end of values suggested by [Imbens and Rubin \(2015\)](#). The results for all cause mortality are notably smaller than our preferred estimates (and less precise). Even with a more conservative trim, we again find decreases in the causes of mortality most likely to be influenced by the ACA Medicaid expansion. The results for amenable causes are slightly smaller than our preferred estimates but remain statistically significant with and without controls.

We also explore how results change when we vary the penalty parameter during the double lasso selection procedure. Appendix Figure A.15 shows results for all-cause mortality across several values of the penalty. As we increase the penalty, fewer variables are selected for inclusion into the propensity score model. A penalty of zero corresponds to all variables selected for inclusion in the model while the largest value, “limit,” corresponds to zero variables selected for inclusion. The optimal penalty is labeled “205*” and corresponds to our main result for all-cause mortality. As we increase the penalty level, the estimated mortality effects becoming small and statistically insignificant. This pattern suggests that as fewer variables are selected for inclusion, the corresponding esti-

mated propensity scores become more noisy. As a result, the re-weighting adjustment does less to correct for pre-existing differences among treatment and control counties and reduces our power to detect mortality effects.²²

5.2. Permutation tests

In addition to investigating how our mortality estimates change across various alternative specifications, we also conduct a permutation test to investigate the likelihood that our model is identifying the causal effects of the ACA Medicaid expansion on mortality or whether the effects we find are simply due to chance as a result of potentially confounding issues such as model mis-specification or lack of power. The permutation test essentially compares our preferred estimates to a distribution of pseudo-treatment effects.

To conduct the permutation test, we first shift the sample period back four years and include the time-period from 2005–2013. We then randomly assign treatment to 27 states, the same number of states that expanded Medicaid in 2014. States that expanded after the first half of 2014 are again excluded from the sample. The remainder of the states make up the control group. After the random assignment to treatment, we follow the same estimation procedure described in Section 3. Specifically, we estimate propensity scores using variables selected by double lasso, i.e. variables that predict the mortality rate for adults ages 20–64 and variables that predict treatment status outcomes. We then estimate the effect of treatment, which begins in 2009 for each treated county, on all-cause and amenable rates of mortality by weighting the regressions by the county population 20 to 64 years of age multiplied by $T + (1 - T) \times \frac{p}{(1-p)}$. We repeat this procedure 1000 times and compare the pseudo-treatment effects to our preferred estimates.

The distribution of results from the permutation test is illustrated in Figure A.18. The figures show binned histograms with our preferred estimates represented by the dashed vertical lines. The 5th and 95th percentiles of the distributions are represented by the solid vertical lines. For all-cause and amenable mortality, our preferred estimates are in the extreme left tail of the distribution of pseudo-treatment effects and fall well outside of the 5th and 95th percentiles. This pattern is similar when we compare

²² We considered adding a state level synthetic control analysis. However, several considerations lead us to not include this analysis in the paper. First, our method allows us to estimate effects when treatment occurs across many units. The synthetic control method was designed for settings that include only one treatment group and many possible controls; the econometric literature on the many-treatment groups case is in development (e.g. [Doudchenko and Imbens \(2016\)](#) and [Robbins et al. \(2017\)](#)) and there is no agreed-upon inference theory for these methods (for recent developments, see [Li \(2017\)](#) and [Chernozhukov et al. \(2018\)](#)). Second, the county-level analysis offers far greater scope for heterogeneity analysis, which is both informative about mechanisms and an important source of robustness. Finally, when we estimated synthetic control models using state-level mortality rates, the results, though consistent with our main analysis, appear less robust, a result due to poor matching of treatment states to the relatively small number of states in the control group ([Ben-Michael et al., 2018](#)).

the estimated *t*-statistics to the distribution of pseudo-treatment *t*-statistics, as shown in Figure A.19. Table A.6 presents summary statistics from the test. For each cause of death, the mean treatment effects are small and generally positive in sign. There are also few estimates that are statistically significant at the conventional 5 percent level, the majority of which are *greater* than zero. The results of this test provide strong evidence that the mortality effects we estimate for the ACA Medicaid expansion are very unlikely to be due to chance.

5.3. Investigating the importance of leverage and outliers

We perform two additional checks that supplement the permutation analysis discussed above. First, we investigate the concentration of leverage in our analysis, i.e. the degree to which individual observations on right-hand side variables take on extreme values and are influential in the regression analysis (Young, 2018). A concentration of leverage in a few observations makes coefficients and standard errors extremely volatile, as their value becomes dependent upon the realization of a small number of residuals. To test whether our results are contingent on concentrated leverage, we first estimate the difference-in-differences model for all-cause and amenable mortality (with and without controls) and calculate the average residuals for each state. In each specification, we then drop the state that has the highest average residual and re-ran the model. Second, we conducted a “leave-one-out” analysis, where we remove each state, one at a time, from the regression analysis. In both tests, we find consistent effect sizes to our preferred estimates, as illustrated in Appendix Table A.7 and Appendix Figure A.20.

5.4. Triple difference with the elderly population

Estimating unbiased causal effects using a difference-in-differences design requires that treatment and control units would have followed parallel trends in the absence of treatment. In our setting, we have shown expansion and control counties did not follow parallel trends in mortality prior to the expansions becoming effective in 2014, which suggests that the unadjusted difference-in-difference would be biased. We have employed the commonly-used method of propensity score re-weighting in combination with model selection techniques from machine learning to address this issue. Our strategy has been effective at constructing a balanced treatment and control group, as measured by parallel pre-trends reported in Figure 1. An alternative strategy would have been to employ a triple-difference strategy using a separate control group that is unlikely to be affected by the Medicaid expansion. BHNS follow Finkelstein and McKnight (2008) in using ages 65 to 74 as an additional control group.

To assess this alternative design, we follow BHNS and estimate a triple-difference model using mortality among adults ages 55 to 74 years of age with our re-weighted sample of counties. An advantage of the triple-difference model is that it allows us to include state-by-year effects. To make

comparisons across groups with different underlying risks we define mortality as $\log\left(\frac{\text{deaths}}{100,000\text{people}} + 1\right)$.

The underlying assumption of this design is that changes in the relative outcomes for the elderly and the non-elderly would evolve similarly in expansion and non-expansion states in the absence of expansion. Unfortunately, even after reweighting the sample using our propensity score model, we find evidence of above-trend mortality in the pre-period in the expansion states for the above-65 age group (see Appendix Table A.16).²³ Although the presence of pre-trends in the elderly group suggests this control group is problematic, when we (effectively) average these pre-trends by the inclusion of the interacted fixed effects, there is less evidence of bias (Appendix Figure A.17). The point estimates shown in Appendix Table A.8 imply just above a 1 percent reduction in all-cause and amenable causes of mortality for adults 55 to 64 years of age compared to the above-65 age group.²⁴ Thus, the DDD analysis provides evidence of mortality reductions that are consistent with our main findings that the Medicaid expansions reduced mortality for the near-elderly population, albeit by a smaller magnitude due to the pre-trends.

6. Social welfare gains

We consider two scenarios that provide us with annual welfare gains associated with the estimated mortality reductions. This exercise compares the estimated annual lives saved, put in dollar terms using standard VSL techniques, to the value of annual total Medicaid expenditures associated with the expansion. In our primary scenario, we focus on amenable causes of mortality among people 55 to 64 years of age, the group where we find the strongest evidence of a mortality response. Since the amenable-cause mortality reductions in this age range do not account for the total estimated effect, however, we interpret this welfare estimate as a lower bound. We provide a second estimate using the all-cause mortality reduction for adults 20 to 64 years of age. It is important to note that our calculations consider only welfare gains from the estimated mortality reductions and do not represent the total welfare gains of the Medicaid expansions. In particular, we do not consider potential welfare gains associated with increases in indi-

²³ In addition to the pre-trends, additional evidence suggests the elderly population may themselves have been impacted by the ACA Medicaid expansion. Specifically, we examine changes in household composition among adults 65 years of age and older who live with a Medicaid recipient between 20 and 64 using data from the American Community Survey. Our findings reveal the percentage of elderly living with a Medicaid recipient increases disproportionately in expansion states after 2014 by 2.3 percentage points. Living together (or in close proximity) is likely the most important channel that connects finances of the newly insured Medicaid population to the elderly. While the full implications of this result extend beyond the scope of our paper, it is evidence that the expansions may have important financial implications for elderly, potentially leading to health improvements that are consistent with studies of earlier Medicaid reforms (e.g. Sommers et al. (2012)).

²⁴ We reran the DDD analysis by estimating propensity scores using all-cause, amenable, and non-amenable death rates for the above-55 sample and found similar results.

Table 4
Estimated welfare gains.

Welfare projections	Mortality reduction per county	Expansion counties	VSLY (\$)	Total gains (\$)
<i>Ages 55–64 Amenable</i>	3.44	1,252	\$4,867,840	\$20,971,877,858
<i>Ages 20–64 All-Cause</i>	9.64	1,252	\$8,430,980	\$101,800,213,514

Notes: VSL values calculated as using estimates from [Aldy and Viscusi \(2008\)](#). All monetary values are reported in 2017 dollars. *Ages 55–64 Amenable* captures only amenable-cause mortality reductions among the population aged 55–64 years. *Ages 20–64 All-Cause* calculates welfare-gains using the all-cause mortality estimate of 3.60 percent for the population 20 to 64 years of age.

vidual or household financial stability, labor supply, and reductions in costly social activity such as crime.

To obtain the primary welfare estimate, we first calculate a “per-county” mortality reduction by multiplying the estimated amenable-cause percentage response in [Table 2](#) for the 55-to-64 age-group by the relevant baseline mortality rate and population (in 100,000s) for the same age-group in expansion counties. We then multiply this value by the total number of expansion counties, 1,252, and by age-specific VSL estimates gathered from [Aldy and Viscusi \(2008\)](#). The procedure for the secondary estimate follows the same approach except that we calculate the “per-county” mortality reduction using the estimated all-cause percentage response of 3.60 percent and the population of individuals 20 to 64 years of age. The VSL estimate for the 20-to-64 category consists of the age-specific VSL estimates, weighted by the ratio of the age-specific mortality rate relative to the average.

Results from the welfare calculations are presented in [Table 4](#). We estimate annual welfare gains (in 2017 dollars) at between \$20.9 and \$102 billion. These estimates equate to between 30 and 145 percent of the \$70 billion in expenditures associated with the Medicaid expansion in 2017 ([Center for Medicare and Medicaid Services, 2017](#)). Given that an estimated 60 percent of Medicaid expenditures are transfers to implicit insurance providers, as opposed to new costs to society, our estimates imply the mortality-related savings may cover the entire net cost of the expansion ([Finkelstein et al., 2016](#)). This calculation sets aside several known issues in valuing mortality reductions. For example, the welfare benefits of mortality reduction could be adjusted for social welfare weights and VSL specific to the target population (see [Finkelstein et al. \(2016\)](#) for longer discussion), as well as harvesting effects (see [Deryugina et al. \(2016\)](#)). The calculation also does not take into account quality of life improvements from changes in morbidity. Nevertheless, even if we were to discount heavily the welfare gains from mortality, our point estimates would still imply that mortality reduction can significantly impact the program’s cost effectiveness.

7. Conclusion

The results we report in this paper add to a growing body of literature on the effects of public health insurance, Medicaid, and the ACA on health outcomes. We find that the ACA Medicaid expansion reduced mortality in the portion of the population that is between 20 and 64 years of age by 11.36 deaths per 100,000, a 3.60 percent reduction, within the first four years of the reform. Considering that the Medicaid expansion covered over 10 percent of the population

in these states, and induced a 4.15 percent net increase in insurance coverage, these estimates imply that the program likely affected mortality through channels beyond the access to healthcare among newly-insured individuals. Improvements in welfare due to the mortality responses we find may cover all annual expenditures associated with the Medicaid expansion.

We believe the estimates derived from the ACA Medicaid expansion provide us with several important advantages relative to the quasi-experiments available to previous researchers. First and most fundamentally, the ACA expansion affected a much larger share of the population than earlier policy reforms. Previous studies of Medicaid expansions have examined at most three state-level reforms. The preeminent study on the effects of Medicaid is the OHIE, the only large-scale randomization of access to Medicaid; however, OHIE lacks sufficient statistical power to detect mortality effects of the magnitude we document here ([Baicker and Finkelstein, 2011](#); [Finkelstein et al., 2012](#)). Second, the nationwide treatment and other details of the ACA reduce the likelihood that concurrent changes in state policy and mortality trends bias the results. Previous reforms were initiated by individual states and may have been accompanied by other changes in policy, or were motivated by specific mortality trends ([Kaestner, 2012](#)). In contrast, the federal government covered 100 percent of the initial costs of Medicaid expansion (and promised to cover 90 percent of the costs starting in 2020), greatly reducing the role of state health and budgetary conditions in the decision to expand. Third, we can conduct several important secondary analyses and robustness checks that do not appear in the previous work. Most notably, we assess pre-reform trends and outcomes in an event study framework, and examine the statistical performance of the model using permutation analysis. Finally, our study serves to update analyses of earlier Medicaid reforms to account for the changing mortality environment. A substantial portion, 20 percent, of the effects of early-2000s Medicaid expansions on mortality came through reductions in HIV-related mortality ([Sommers, 2017](#)). Since then, medical innovations have greatly reduced mortality due to HIV, while other causes of death have risen, in both relative and absolute terms ([Okie, 2010](#); [Phillips, 2014](#); [Case and Deaton, 2015](#)). Our findings suggest that the ACA Medicaid expansion resulted in meaningful reductions in mortality, despite the changing mortality environment.

Authors’ contributions

Borgschulte: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources,

Writing – Original Draft, Writing – Review and Editing, Visualization, Supervision, Project administration

Vogler: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Writing – Original Draft, Writing – Review and Editing, Visualization, Supervision, Project administration

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.jhealeco.2020.102333>.

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